21/11/19

METRIC SPACE

Norms & Banach Spaces

Norm: Let X be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A norm on X is a function $\|\cdot\| : X \to \mathbb{R}$ such that $\forall x, y \in X$ and all scalars c, we have (a) $\|x\| \geq 0$ and $\|x\| = 0 \Leftrightarrow x = 0$

(b) $\|cx\| = |c|\|x\|$

(c) $\|x+y\| \leq \|x\| + \|y\|$

A vector space X together with a norm is called a normed vector space, a normed linear space or simply a normed space.

$\|x\| \to$ length of a vector x

$\|x-y\| \to$ distance between two vectors x and y

~~E.g.~~: The Induced Metric

If X is a normed space, then it follows directly from the definition of a norm that $d(x,y) = \|x-y\|$, $\forall x, y \in X$ defines a metric on X. This is called the metric on X induced from the norm $\|\cdot\|$.

Properties of Norm

If X is a normed space, then the following statements hold -

(i) $\big|\|x\| - \|y\|\big| \leq \|x-y\|$;

(ii) convergence $\Rightarrow$ cauchy i.e. $x_n \to x \Rightarrow \{x_n\}$ is a Cauchy sequence;

[converse in general is not true]

(iii) continuity of the norm: $x_n \to x \Rightarrow \|x_n\| \to \|x\|$.

Banach Spaces

A normed space X is a Banach space if it is complete, i.e., if every cauchy sequence in X converges to an element in X.

E.g.: The set of real no.s $\mathbb{R}$ is complete w.r.t. the absolute values using real scalars, and likewise, the complex plane $\mathbb{C}$ is complete w.r.t. the absolute value using complex scalars.

More generally, $\mathbb{R}^n$ and $\mathbb{C}^n$ are Banach spaces w.r.t. Euclidean norm.

For $x \in \mathbb{R}^n$, $\|x\|_2 = \|(x_1, x_2, \dots, x_n)\| = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}$

$d_2(x,y) = \|x-y\|_2 = \left(\sum_{i=1}^{n} (x_i - y_i)^2\right)^{1/2}$

Note :-

Every finite dimensional vector space V is complete w.r.t. any norm that we place on V.

Example: (l^p - Space)

The space l^p of complex sequences with the norm $\|x\|_p := \left(\sum_{k=1}^{\infty} |x_k|^p\right)^{1/p}$, $x = (x_k) \in l^p$. It can be shown that l^p is a vector ~~the~~ space and the norm $\|\cdot\|_p$ defined above is indeed a norm on l^p.

$\therefore$ l^p is a normed space.

Theorem : Completeness of l^2

Let $(a_n) \in l^2$ be cauchy where $a_n = (\alpha_1^{(n)}, \alpha_2^{(n)}, \dots)$ be an arbitrary sequence.

Let $\varepsilon > 0$ be given.

Then, $\exists$ M such that $n, m \geq M$ for which -

$\|a_n - a_m\|^2 = \sum_{k=1}^{\infty} |\alpha_k^{(n)} - \alpha_k^{(m)}|^2 < \varepsilon^2$.

For any $k = 1, 2, 3, \dots$ and $m, n \geq M$, we can write

$|\alpha_k^{(n)} - \alpha_k^{(m)}| < \varepsilon$.

$\Rightarrow$ $(\alpha_k(n))_k$ is a Cauchy sequence of complex no.s.

Let $\alpha_k = \lim_{n \to \infty} \alpha_k^{(n)}$ and $a = (\alpha_1, \alpha_2, \dots)$.

We will now show that $a \in l^2$ and $a_n \to a$.

We have, for any $n \geq M$, $\|a_n - a\|^2 \leq \varepsilon^2$

So, by Minkowski's Inequality, we can write

$\|a\| = \|a - a_M + a_M\| \leq \|a - a_M\| + \|a_M\| < \infty$

$\Rightarrow a \in l^2$

Also, $\lim_{n\to\infty} \|a_n - a\| = 0$

$\Rightarrow a_n \to a$

$\therefore (l^2, \|\cdot\|_2)$ is a Banach space.

Proved.

The space $C_b(X)$

Let X be a metric space and $C_b(X)$ be a space of all bounded and continuous functions $f : X \to \mathbb{C}(\mathbb{R})$. The uniform norm of a function $f \in C_b(X)$ is defined as ~~Δ~~ $\|f\|_u := \sup_{x\in X} |f(x)|$. This is a norm on $C_b(X)$.

If $f_n, f \in C_b(X)$ and $\lim_{n\to\infty} \|f_n - f\| = \lim_{n\to\infty} \left(\sup_{x\in X} |f_n(x) - f(x)|\right) = 0$,

then we say that f_n converges uniformly to f.

Theorem

Let $\{f_n\}$ be the Cauchy sequence in $C_b(X)$ w.r.t. $\|\cdot\|_u$, then $\exists$ a function $f \in C_b(X)$ such that $f_n \xrightarrow{u.c.} f$. Consequently, $C_b(X)$ is a Banach space w.r.t. $\|\cdot\|_u$.

28/11/19

Metric Space

Theorem

$C_b(X)$ is a Banach space w.r.t. $\|\cdot\|_u$

Proof :-

Suppose, $\{f_n\}$ be a uniformly cauchy sequence in $C_b(X)$. If we fix any particular point $x \in X$, then ~~$|f_m(x) - f_n(x$~~

$|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_u \;\forall m, n.$

$\Rightarrow \{f_n(x)\}$ is cauchy sequence of scalars.

$\because \mathbb{C}$ is complete, this cauchy sequence of scalars must converge.

Define $f(x) = \lim_{n\to\infty} f_n(x),\ x \in X$

Now, choose $\varepsilon > 0$, then $\exists\ N$ such that $\|f_m - f_n\| < \varepsilon, \forall m, n > N$.

If $n > N$, then for every $x \in X$, we have $|f(x) - f_n(x)| = \lim_{m\to\infty} |f_m(x) - f_n(x)|$

$\leq \|f_m - f_n\|$

$< \varepsilon$

Hence, $\|f - f_n\|_u \leq \varepsilon,\ \forall n > N$

This shows that $f_n \to f$ uniformly as $n \to \infty$

Consequently, $f \in C_b(X)$.

Thus, every uniformly cauchy sequence in $C_b(X)$ converges in $C_b(X)$.

$\therefore$ We have shown that $C_b(X)$ is complete w.r.t. uniform norm.

Thus, $C_b(X)$ is Banach space w.r.t. uniform norm.

Weierstrass Approximation Theorem

This theorem is an important result about $C_b(X)$ when $X = [a, b]$ is a finite closed interval in the real line. This theorem tells us that the set of polynomial functions is a dense subset of $C[a,b]$.

Statement :- Given a function $f \in C[a,b]$ and for $\varepsilon > 0$, $\exists$ some polynomial $p(x) = \sum_{k=0}^{N} c_k x^k$ such that $\|f - p\|_u = \sup_{x \in [a,b]} |f(x) - p(x)| < \varepsilon$.

Theorem

A closed vector subspace of a Banach space is a Banach space.

Proof :-

Let F be a closed vector subspace of a Banach space E. Let $\{x_n\}$ be a Cauchy sequence in F. Consequently, $\{x_n\}$ is a Cauchy sequence in E and $x_n \to x \in E$.

$\Rightarrow x \in F$ [$\because F$ is closed]

Theorem

Every finite dimensional normed linear space is a Banach space.

Theorem

Any two norms on a finite dimensional normed space are equivalent.

03/12/19

Contraction Mapping and Banach Fixed Point Theorem

Let (X, d) be a metric space. A mapping $T: X \to X$ is said to be contraction mapping on X if $\exists\ \alpha \in \mathbb{R}$ with $0<\alpha<1$ such that $d(Tx, Ty) \le \alpha\, d(x, y)$. The contraction mapping is uniformly continuous on X.

Examples :-

(i) Let $f: [a,b] \to [a,b]$ be a differential function with $|f'(x)| \le k$, $\forall x \in [a,b]$, $\forall k<1$. Then, f is a contraction mapping.

(ii) $Tx = x^2$, $0 \le x \le \frac{1}{n+1}$, $n \geq 1$; then T is a contraction mapping on $[0, 1/_{n+1}]$.

Proof:
$$d(Tx, Ty) = x^2 - y^2 = (x+y)(x-y)$$
$$< \frac{2(x-y)}{n+1}$$
$$= \frac{2}{n+1} d(x,y)$$

Here, $\alpha = \frac{2}{n+1}$. If $n \geq 1$ then $n+1 \geq 2$
$$\Rightarrow \frac{2}{n+1} \le 1$$
$$\Rightarrow \alpha \le 1.$$

$\therefore$ ~~Tx~~ T is a contraction mapping.

(iii) Let $f: [0,1] \to [0,1]$ defined by $f(x) = \frac{1}{x+1}$. Is f a contraction mapping on $[0,1]$.

Ans.:
$$d(f(x), f(y)) = \frac{1}{x+1} - \frac{1}{y+1}$$
$$= \frac{y+1-x-1}{(x+1)(y+1)}$$
$$= \frac{y-x}{(x+1)(y+1)}$$
$$\le -\frac{1}{4} d(x,y)$$

Def^n: Fixed Point

A point $x \in X$ is called a fixed point if the mapping T is defined as $T(x) = x$.

A mapping has no or one or more than one fixed points, in general.

Banach Fixed Point Theorem / Contraction Mapping Principle

If (X, d) be a complete metric space and $T: X \to X$ be a contraction mapping, then T has a unique fixed point in X.

(uniformly continuous)

Proof :-

Let $x_0 \in X$ and $\{x_n\}$ be the sequence of iteratives defined by $x_{n+1} = Tx_n$, $\forall n \geq 1$.

We will show that $\{x_n\}$ is a Cauchy sequence in X.

Now, $\forall p = 1, 2, 3, \ldots$ we have -

$$d(x_{p+1}, x_p) = d(Tx_p, Tx_{p-1})$$
$$\leq \alpha\, d(x_p, x_{p-1}) \quad \text{—— ①}$$

where $0 < \alpha < 1$ is such that $d(Tx, Ty) \leq \alpha\, d(x, y)$ [$\because T$ is a contraction mapping].

$\therefore$ For m, n where $m \geq n \geq 1$, we can write -

$$\cancel{d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \ldots + d(x_{n+1}, x_n)}$$
$$\cancel{\leq \alpha\, d(x_{m-1}, x_{m-2}) + \ldots + \alpha d(x}$$

Now, applying ① repeatedly, we can write -

$$d(x_{p+1}, x_p) \leq \alpha\, d(x_p, x_{p-1}) \leq \alpha^2 d(x_{p-1}, x_{p-2}).$$
$$\leq \ldots$$
$$\leq \alpha^p d(x_1, x_0) \quad \text{—— ②}$$

Let $m, n \in \mathbb{N}$ with $m \leq n \leq 1$, then from triangle inequality -

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + \ldots + d(x_{n+1}, x_n)$$
$$\leq \alpha^{m-1} d(x_1, x_0) + \ldots + \alpha^n d(x_1, x_0) \quad [\text{from ②}]$$
$$= \alpha^n \left[\alpha^{m-n-1} + \alpha^{m-n-2} + \ldots + 1\right] d(x_1, x_0)$$

$$= \alpha^{n}\left[\sum_{k=0}^{m-n-1} \alpha^{k}\right] d(x_1, x_0)$$

$$= \alpha^{n}\sum_{k=0}^{\infty}\alpha^{k} d(x_1, x_0)$$

$$\leq \frac{\alpha^{n}}{1-\alpha} d(x_1, x_0)$$

But $\lim_{n\to\infty} \alpha^{n} = 0$, $\therefore$ $\{x_n\}$ is a cauchy sequence in X.

$\because$ X is complete, so the cauchy sequence $\{x_n\}$ converges to some point in X.

Let $\lim_{n\to\infty} x_n = x^*$, $x^* \in X$.

We have to show that $Tx^* = x^*$.

$\because$ T is a contraction mapping, $\therefore$ T is uniformly continuous in X and hence, continuous.

$$\therefore Tx^* = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} x_{n+1} = x^*$$

$\therefore$ x^* is a fixed point.

Now, we have to show that x^* is unique.

Let $y^* \in X$ be another fixed point of T.

Then, $Ty^* = y^*$.

Now, $0 \leq d(x^*, y^*) = d(Tx^*, Ty^*)$
$\leq \alpha d(x^*, y^*)$
$< d(x^*, y^*)$ $[\because 0<\alpha<1]$.

$\Rightarrow d(x^*, y^*) = 0$

$\Rightarrow x^* = y^*$

$\therefore$ T has a unique fixed point.

Hence, proved.

Applications of Banach's Fixed Point Theorem in Linear Algebra

Let $y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i$, $i = 1, 2, \ldots, n$ can be thought of as a mapping $T: \mathbb{R}^n \to \mathbb{R}^n$ defined by $y = Tx = ax + b$, where $x = (x_1, x_2, \ldots, x_n)^T$, $y = (y_1, \ldots, y_n)^T$, $b = (b_1, \ldots, b_n)^T$. We now determine sufficient condition for T to be a contraction for T under different choices of metric on $\mathbb{R}^n$.

Case 1: under the metric defined on $\mathbb{R}^n$ by

$$d_2(x, y) = \left[\sum_{i=1}^{n} (x_i - y_i)^2\right]^{1/2}, \; x, y \in \mathbb{R}^2$$

Now,

$$d_2^2(y, \tilde{y}) = \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2$$

$$= \sum_i \left[\sum_j a_{ij}(x_j - \tilde{x}_j)\right]^2$$

$$\leq \sum_i \left[\left(\sum_j a_{ij}^2\right)\left(\sum_j (x_j - \tilde{x}_j)^2\right)\right]$$

$$\Rightarrow d_2^2(y, \tilde{y}) \leq \left(\sum_i \sum_j a_{ij}^2\right) d_2^2(x, \tilde{x}).$$

Thus, if $\left(\sum_i \sum_j a_{ij}^2\right) < 1$, for each i then T is a contraction mapping on $(\mathbb{R}^n, d_2)$.

Case 2: Consider the metric $d_\infty(x, y) = \max_i |x_i - y_i|$, $x, y \in \mathbb{R}^n$.

$$d_\infty(y, \tilde{y}) = \max_i |y_i - \tilde{y}_i|$$

$$= \max_i \left|\sum_j a_{ij}(x_j - \tilde{x}_j)\right|$$

$$\leq \max_i \sum_j |a_{ij}||x_j - \tilde{x}_j|$$

$$\leq \left(\max_i \sum_j |a_{ij}|\right)\left(\max_j |x_j - \tilde{x}_j|\right)$$

$$= \left(\max_i \sum_j |a_{ij}|\right) d_\infty(x, \tilde{x})$$

Thus, if $\max_i \sum_j |a_{ij}| < 1$, for each i, then T is a contraction mapping on $(\mathbb{R}^n, d_\infty)$, which is certainly less condition than that in case 1.

Case 3: Consider the usual metric on $d_1(x,y) = \sum_i |x_i - y_i|$.

Suff. condition : $\sum_i |a_{ij}| < 1$.

In all the cases $(\mathbb{R}^n, d_1 / d_2 / d_\infty)$ is complete. So, T has a fixed point that gives the unique solution of the linear system.

$$d_1(y, \tilde{y}) = \sum_i |y_i - \tilde{y}_i|$$

$$= \sum_i \left| \sum_j a_{ij} x_j - \sum_j a_{ij} \tilde{x}_j \right|$$

$$= \sum_i \left| \sum_j a_{ij} (x_j - \tilde{x}_j) \right|$$

$$\cancel{\leq \sum_i \sum_j |a_{ij}| |x} \leq \sum_i \left| \sum_j a_{ij} \right| \left| \sum_j (x_j - \tilde{x}_j) \right|$$

$$= \sum_i \left| \sum_j a_{ij} \right| d_1(x, \tilde{x})$$

$$= \left(\sum_i \sum_j |a_{ij}| \right) d_1(x, \tilde{x})$$

If $\sum_i \sum_j |a_{ij}| < 1$ for each i, then T is a contraction mapping on $(\mathbb{R}^n, d_1)$.